The 'hello world' of Monte Carlo

Part 1: The $e^+e^- \to q\bar{q}$ differential and total cross section

We will consider the process

$$e^+e^- \to \gamma^* \to q\bar{q}.$$

We work under the assumption that only the photon-exchange diagram is relevant. Furthermore, we assume that we are in the centre-of-mass (CM) frame of the colliding electrons (i.e. they are back-to-back) and assume negligible particle masses. The CM energy is denoted by $\sqrt{s}$. We denote the momenta of incoming electron-positron pair by p_1 and p_2, and those of the outgoing quarks by q_1 and q_2.

During the lecture, we saw that the matrix element is given by

$$iM_{e^+e-\to q\bar{q}} = \frac{ie^2 Q_q}{(q_1+q_2)^2}\bar{u}(p_1)\gamma^\mu v(p_2)\,\bar{v}(q_2)\gamma_\mu u(q_1)\,,$$

with Q_q the charge of the quark, and e the electromagnetic coupling.

1. Using Dirac algebra, show that the squared matrix element is equal to $|M|^2 = e^4 Q_q^2 N_c(1+\cos^2\theta)$, where θ is the angle between the outgoing quark and the electron (in the CM frame), and the number of colours is denoted by N_c. Remember that all particles are taken to be massless. You will need
$$\mathrm{Tr}[\gamma^\mu\gamma^\nu\gamma^\sigma\gamma^\rho] = 4\left(g^{\mu\nu}g^{\sigma\rho} - g^{\mu\sigma}g^{\nu\rho} + g^{\mu\rho}g^{\nu\sigma}\right).$$
2. Draw the distribution in θ of the squared matrix element from $\theta = 0$ to $\theta = \pi$.
3. Starting from the defining expression of the Lorentz-invariant phase space for two particles, i.e.
$$\int d\Phi_2 = \int \frac{d^3q_1}{2E_q(2\pi)^3}\frac{d^3q_2}{2E_{\bar{q}}(2\pi)^3}(2\pi)^4\delta^4(p_1+p_2-q_1-q_2)\,,$$
show that
$$\int d\Phi_2 = \frac{1}{8\pi}\int d\cos\theta\,.$$
Why are we allowed to integrate over the azimuthal angle ϕ?

Taken together, we find that the total cross section for $e^+e^- \to \gamma^* \to q\bar{q}$ becomes

$$\sigma(e^+e^- \to \gamma^* \to q\bar{q}) = \frac{4\pi\alpha_{\mathrm{EM}}^2}{3s}N_c\sum Q_q^2\theta(m_q^2 > s)\,.$$

These results will be used to write and validate our Monte Carlo.

Part 2: Our first event generator

Instead of computing the cross section analytically, we now want to set up a program that calculates it numerically. We will assume you work in python, but you are free to use any other language you are comfortable with. Random numbers between 0 and 1 may be obtained using

```
import random
r = random.uniform(0,1)
```

1. Using the fact that
$$\int_{x_{\min}}^{x} dx f(x) = r\int_{x_{\min}}^{x_{\max}} dx f(x)$$
write a program returns a sample of x values that is distributed according to $f(x) = 1 + x^2$ with $x_{\min} = -1$ and $x_{\max} = 1$. Hint: one can use a numerical bisection method to find the solution to an equation of the form $g(x) = 0$, or you can solve the equation analytically. Plot the resulting distribution in x. Does it agree with your expectation (part 1, item 2)? How many samples does one need to get a satisfactory distribution?

2. Write a program that returns a sample of x values distributed according to $f(x) = 1 + x^2$ using an accept-and-reject method. Which of (1) and (2) is faster?

3. Numerically compute the integral of
$$\int_{-1}^{1} dx(1+x^2)\,.$$

4. How can you compute the integral and at the same time obtain a set of x values distributed according to $f(x)$? Write a program that does this.

5. Defining now all relevant constants, which we set to $\alpha_{\rm EM} = 1/137$, $N_c = 3$, $Q_u = Q_c = 2/3$, $Q_d = Q_s = Q_b = -1/3$, plot the total cross section of $e^+e^- \to \gamma^* \to q\bar{q}$ in pb as a function of the scattering energy s for $\sqrt{s} = 1 - 100$ GeV.

You've now written an LO event generator that computes σ and $d\sigma/d\cos\theta$!

Part 3: Our first NLO event generator

At NLO we have to deal with the soft and collinear singularities that pop up. In this exercise, we will consider a toy model, and compute the NLO cross section numerically using two different techniques: *subtraction* and *slicing*. This toy model only has one divergence, and hides some of the complexities one has to deal with when calculating an actual NLO correction to e.g. the $e^+e^- \to q\bar{q}$ process considered before.

Consider a system radiating off massless particles that carry away a collective energy of x. At Born level, we do not have any emissions, and $x = 0$. At NLO we need to take into account loop corrections and real emission corrections. We define the following differential cross sections for the Born, Virtual and Real contributions, after dimensional regularisation to compute the loop integrals:

$$\begin{aligned}
\left(\frac{d\sigma}{dx}\right)_{\rm B} &= B\delta(x)\,,\\
\left(\frac{d\sigma}{dx}\right)_{\rm V} &= \alpha\left(\frac{B}{2\epsilon} + V_f\right)\delta(x)\,,\\
\left(\frac{d\sigma}{dx}\right)_{\rm R} &= \alpha\frac{R(x)}{x}\,.
\end{aligned} \tag{1}$$

Here we have introduced a dimensional regulator ϵ, in addition to arbitary constants B and V_f, the coupling α, and a to-be-specified function $R(x)$. To compute the total cross section we need to evaluate

$$\sigma = \lim_{\epsilon\to 0}\int_0^1 dx\, x^{-2\epsilon}\left[\left(\frac{d\sigma}{dx}\right)_{\rm B} + \left(\frac{d\sigma}{dx}\right)_{\rm V} + \left(\frac{d\sigma}{dx}\right)_{\rm R}\right] \equiv \sigma_B + \sigma_V + \sigma_R\,. \tag{2}$$

Clearly, we cannot dump this on a computer as-is: there is a limit of $\epsilon \to 0$ to be taken care of.

1. Compute the contributions from the Born and virtual terms, i.e. the term $\sigma_B + \sigma_V$.

2. *Subtraction*: For the subtraction technique, we write the real contribution as
$$\sigma_R = \alpha\left[\int_0^1 dx\, x^{-2\epsilon}\frac{R(x)-R(0)}{x} + R(0)\int_0^1 dx\, x^{-1-2\epsilon}\right].$$

   (a) Evaluate the rightmost integral.

   (b) What must be the condition on $R(0)$ such that the divergence in the virtual contribution cancels?

   (c) What must be the condition on $R(x)$ such that the first integral is integrable in 4 dimensions ($\epsilon = 0$)?

   (d) Write an expression for the total cross section σ that can be evaluated directly in 4 dimensions.

3. *Slicing*: In slicing, we introduce a cutoff parameter $\delta \ll 1$ to write

$$\int_0^1 dx = \int_0^\delta dx + \int_\delta^1 dx\,.$$

   (a) Approximate $R(x) \to R(0)$ when $x < \delta$. Compute the contribution from the region $x < \delta$ analytically, and expand in the limit that $\delta \to 0, \epsilon \to 0$ with $\delta \gg \epsilon$. What must be the condition on $R(0)$ to cancel the virtual divergence?

   (b) Write the contribution for $x > \delta$ in a form suitable for numerical integration.

4. *Monte Carlo implementation:* We now pick a simple function satisfying the required properties

$$R(x) = B(1 + x(1-x))\,.$$

With this we obtain a total cross section of

$$\sigma = B\left(1 + \alpha\left(\frac{V_f}{B} + \frac{1}{2}\right)\right).$$

   (a) Convince yourself of this answer in three ways: using Eq. (2) directly, using *subtraction* and using *slicing*.

   (b) Write a Monte Carlo program that computes σ using *subtraction* and *slicing*. Take $B = 2$, $V_f = 1$ and $\alpha = 0.1$ for the numerical evaluation of the cross section. Which method is more stable? Can you pick δ arbitrary small?